

VTK-m: Visualization for the Exascale Era and Beyond

Mark Bolstad*
mbolsta@sandia.gov
Sandia National Laboratories
USA

Kenneth Moreland
David Pugmire
morelandkd@ornl.gov
pugmire@ornl.gov
Oak Ridge National Laboratory
USA

David Rogers
Li-Ta Lo
dhr@lanl.gov
ollie@lanl.gov
Los Alamos National Laboratory
USA

Berk Geveci
berk.geveci@kitware.com
Kitware, Inc.
USA

Hank Childs
hank@uoregon.edu
University of Oregon
USA

Silvio Rizzi
srizzi@anl.gov
Argonne National Laboratory
USA



Figure 1: Datasets rendered with VTK-m. From left to right: a. Temperature field computed by the Nek5000 thermal hydraulics simulator, b. Volume rendering of a simulated supernova explosion, c. Rendering of an idealized Inertial Confinement Fusion (ICF) simulation of a Rayleigh-Taylor instability with two fluids mixing in a spherical geometry, d. Poincaré plot of fusion simulation

ABSTRACT

A recent trend in modern high-performance computing is the increasing use of hybrid architectures, where the vast majority of performance comes from accelerators. Modern accelerators are based on Graphics Processing Units (GPU) that contain many low power cores that in their aggregate provides an extremely high computation rate. Current and future CPU processors are requiring more explicit parallelism as each successive version of the hardware packs in more cores, and technologies like hyperthreading and vector operations require even more parallel processing to leverage each core's full potential. As an example, the Frontier supercomputer installed at Oak Ridge National Laboratories recently hit a record breaking 1.1 exaflops¹ on the LINPACK HPC benchmark [Shoemaker 2022]. The system contains 37632 AMD MI250x GPUs which requires more than half a billion threads to keep the system fully utilized [Khizeran 2022].

VTK-m is a toolkit of scientific visualization algorithms for these emerging processor architectures. VTK-m supports the fine-grained concurrency for data analysis and visualization algorithms required

to drive extreme scale computing by providing abstract models for data and execution that can be applied to a variety of algorithms across many different processor architectures.

CCS CONCEPTS

- **Computing methodologies** → **Massively parallel algorithms;**
- **Human-centered computing** → **Visualization toolkits.**

KEYWORDS

high-performance computing, scientific visualization

ACM Reference Format:

Mark Bolstad, Kenneth Moreland, David Pugmire, David Rogers, Li-Ta Lo, Berk Geveci, Hank Childs, and Silvio Rizzi. 2023. VTK-m: Visualization for the Exascale Era and Beyond. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH '23 Talks)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3587421.3595466>

1 INTRODUCTION

VTK-m, whose name was chosen to invoke a familiarity with the Visualization Toolkit (VTK), came from different origins. The US Department of Energy (DoE) high-performance computing (HPC) community predicted early on that leadership class facilities would be transitioning to heavily threaded processors and that we would need a significant change to the visualization algorithms and software [Ahern 2011]. Consequently, researchers at the DoE national laboratories began considering the challenges of visualization on accelerator processors and created three separate toolkits, PISTON [Lo et al. 2012], EAVL [Meredith et al. 2012], and Dax [Moreland

*VTK-m is the product of a large open-source community without whose combined efforts, this paper would not have been possible

¹An exaflop is one quadrillion floating point operations per second

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '23 Talks, August 06–10, 2023, Los Angeles, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0143-6/23/08.

<https://doi.org/10.1145/3587421.3595466>

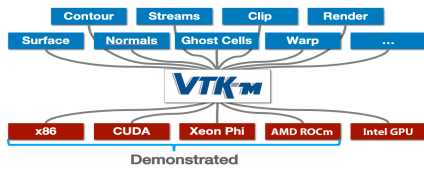


Figure 2: VTK-m abstraction of many-core devices. Intel GPU abstraction layer is under development

et al. 2012], each focusing on a specific aspect of the problem. The three projects were unified under a new code base, with each team continuing to contribute and advance the technologies under the combined VTK-m umbrella.

Through the use of an approach we call MCD³, Meta-Data Parallel Primitives (DPP), Convenience routines, Data management, DPPs, and Devices [Moreland et al. 2021], VTK-m abstracts away the underlying many-core devices allowing developers to focus on the visualization algorithm (see figure 2).

2 PARALLEL PRIMITIVES

As noted in the introduction, VTK-m uses two different types of parallel primitives to achieve performance and reduce developer costs. Meta-DPPs identify what type of data to parallelize over, e.g., parallelizing over cells, points, or elements in an array.

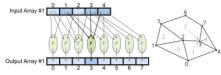


Figure 3: Visit Point with Cells

They provide infrastructure that organizes input data (arrays or data sets) for execution on many-core devices based on the parallelization type (cells, points, elements in an array). An algorithm developer then only needs to provide a functor to carry out the operation for a single instance of the parallelization type, e.g., a single cell. The meta-DPP (in the example of parallelizing over cells, see figure 3) identify which points are incident to a cell and provide their spatial locations and field values to the functor. VTK-m has implemented only five Meta-DPPs, Visit Point With Cells, Visit Cell With Points, Point Neighborhood, Reduce By Key, and Map Field, but VTK-m’s visualization algorithms contain more than eighty instances of these DPPs.

3 PERFORMANCE

A very legitimate question to ask is "With the heavy abstraction, how does VTK-m perform?" In the MCD³ paper [Moreland et al. 2021], a study was run to compare nine individual algorithms implemented in VTK-m against versions from visualization/rendering software (Embree, HAVS, OptiX, VisIt, VTK, Vapor) and direct implementations (CUDA, OpenMP, pthreads, TBB, Thrust) (figure 4). The aggregate performance across all of the tests was 1.14, meaning that across all tests, VTK-m was 14% faster. The overall performance is lowered by the worse performance for the ray-tracing algorithm, due to the highly optimized performance of Embree and OptiX.

VTK-m also operates in-situ with some of the largest scale simulations ever run. In Figure 1 the third image is a visualization of a Rayleigh-Taylor instability run on 16k GPUs of a pre-exascale

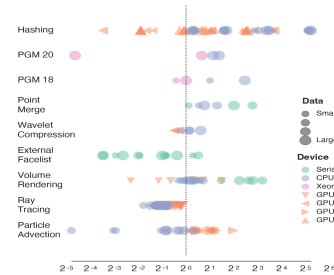


Figure 4: VTK-m performance against hardware-specific comparators for nine algorithms. The horizontal axis represents the ratio in VTK-m performance against its comparator. The glyphs indicate both the hardware type and the normalized size of the data set being processed.

system. Additionally, the image on the right in Figure 1 shows a fusion Poincaré plot that VTK-m accelerated by more than sixty times. At the time of publication, VTK-m will have been used for visualization on some of the largest simulations ever run.

4 CONCLUSION

VTK-m is currently the only software capable of visualizing the largest simulations at scale. Because of its proven capabilities and performance, VTK-m is being integrated into some of the most popular visualization software (VTK, ParaView, and VisIt) as a replacement for their underlying algorithms, thus enabling those packages to visualize at scale.

ACKNOWLEDGMENTS

This research was supported by the Exascale Computing Project (17-SC-20-SC), a collaborative effort of the U.S. Department of Energy Office of Science and the National Nuclear Security Administration.

REFERENCES

- Sean Ahern. 2011. Scientific Discovery at the Exascale: Report from the DOE ASCR 2011 Workshop on Exascale Data Management, Analysis, and Visualization. <http://science.energy.gov/-/media/ascr/pdf/program-documents/docs/Exascale-ASCR-Analysis.pdf>
- Nahren Khizeran. 2022. Hewlett Packard Enterprise ushers in new era with world’s first and fastest exascale supercomputer “Frontier” for the U.S. Department of Energy’s Oak Ridge National Laboratory. <https://www.hpe.com/us/en/newsroom/press-release/2022/05/hewlett-packard-enterprise-ushers-in-new-era-with-worlds-first-and-fastest-exascale-supercomputer-frontier-for-the-us-department-of-energys-oak-ridge-national-laboratory.html>
- Li-ta Lo, Christopher Sewell, and James Ahrens. 2012. PISTON: A Portable Cross-Platform Framework for Data-Parallel Visualization Operators. *Proceedings Eurographics Symposium Parallel Graphics and Visualization (EGPGV)* (2012), 11–20.
- Jeremy S. Meredith, Sean Ahern, Dave Pugmire, and Robert Sisneros. 2012. EAVL: The Extreme-scale Analysis and Visualization Library. *Eurographics Symposium on Parallel Graphics and Visualization* (2012).
- Kenneth Moreland, Brad King, Robert Maynard, and Kwan-Liu Ma. 2012. Flexible Analysis Software for Emerging Architectures. *Proc. SC’12 Companion: High Performance Computing, Networking Storage and Analysis* (2012), 821–826.
- Kenneth Moreland, Robert Maynard, David Pugmire, Abhishek Yenpure, Allison Vacanti, Matthew Larsen, and Hank Childs. 2021. Minimizing Development Costs for Efficient Many-Core Visualization Using MCD3. *Parallel Comput.* (2021).
- Sara Shoemaker. 2022. Frontier supercomputer debuts as world’s fastest, breaking exascale barrier. <https://www.ornl.gov/news/frontier-supercomputer-debuts-worlds-fastest-breaking-exascale-barrier>